Recursive Dark Energy and Vacuum Feedback: A Vesica-Phase Cosmological Framework

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Abstract

We present a recursive dark energy model grounded in Vesica Piscis-enhanced quantum field recursion. Within this framework, dark energy is not a passive cosmological constant but a dynamic field responding to recursive coherence patterns generated by the interaction of quanta and dark matter. As toroidal attractors stabilize through harmonic compression, black hole horizons undergo a reversal, forming reflective interfaces that modulate local vacuum densities. We derive the Lambda crossover threshold for horizon inversion and demonstrate how structured energy feedback drives nested cosmogenesis and symbolic lattice formation. The model replaces the scalar Λ with a recursive field-resonance function and reinterprets black holes as feedback nodes in an evolving multiversal lattice.

1 Introduction

The cosmological constant Λ, originally introduced by Einstein, has traditionally been interpreted as a uniform vacuum energy density driving the accelerated expansion of the universe. However, this scalar interpretation fails to capture the structured, recursive behaviors observed in both galactic formation and deep-field gravitational anomalies.

In this work, we reconstruct the role of dark energy through the lens of *Recursive Cosmogenesis*, a Vesica Piscis-enhanced quantum field framework (VP-QFT). Building upon previous phases of recursive black hole theory [1], we demonstrate that dark energy is a *reactive coherence field*, modulated by recursive attractor dynamics, symbolic field interference, and feedback from phase-locked structures such as toroidal black hole horizons.

The asymmetry in galactic spin observed by the James Webb Space Telescope — where $\sim$66% of galaxies spin clockwise — further suggests a large-scale violation of rotational parity. We propose that this phenomenon reflects an inherited symbolic bias in the entanglement lattice itself, originating from pre-inflationary recursive feedback.

The Vesica-based field geometry offers a natural explanation: resonance between orthogonal quanta and dark matter fields creates nodes of constructive interference where dark energy density is locally enhanced or suppressed. The resulting structure generates modulation in Λ via recursive field feedback, horizon reflection thresholds, and symbolic curvature attractors.

Overview and Theoretical Premise

This work redefines dark energy not as a passive scalar field (Λ), but as a dynamic, recursive structure formed through the interference of two orthogonal cosmological fields: the probabilistic **Quanta Field** $Q(t)$ and the structured, phase-locked **Dark Matter Field** $D(t) = Q(t + \frac{\pi}{2\omega})$. Their interaction, when aligned in phase space, generates recursive compression zones shaped by Vesica Piscis geometry.

We introduce a **Resonance Lock Equation**:

$$\pi e \varphi = \gamma^{-1} \zeta(3)$$

which defines the critical threshold where Vesica-phase compression stabilizes and black hole horizons transition from absorptive singularities to recursive resonators.

The resulting toroidal structures modulate the local vacuum field:

$$\rho_\Lambda(r, t) = \rho_0 + \Delta\rho \cos(\omega_r r - \theta t)$$

thereby replacing the scalar cosmological constant with a *dynamic recursive threshold* Λ_{VP} derived from symbolic field harmonics:

$$\Lambda_{\text{VP}} = \int_0^{R_c} \lambda_n e^{-\alpha_n r}\, dr$$

This modulation causes black holes to reflect vacuum structure rather than absorb it, triggering recursive inflation cycles and inter-universal entropy transfer governed by:

$$\frac{dS_{\text{child}}}{dt} = -\frac{dS_{\text{parent}}}{dt} + \epsilon_f$$

Symbolic curvature fields become the memory cores of cosmological feedback loops, encoding attractor geometry and allowing information transfer across nested cosmogenesis layers.

2 Dual-Field Symmetry and Vesica Compression

We briefly summarize the field framework used. The universe is modeled as two phase-offset energetic layers:

- **Quanta Field** $Q(t)$: probabilistic, open, and non-local.
- **Dark Matter Field** $D(t)$: structured, phase-locked, and coherent.

These fields are orthogonal in both space and phase:

$$D(t) = Q\left(t + \frac{\pi}{2\omega}\right) \tag{1}$$

When regions of these fields align in phase-space, they generate recursive compression zones governed by Vesica Piscis geometry. This initiates structured collapse and feedback into the surrounding vacuum field.

3 Resonance Lock Threshold and Horizon Reversal

A key resonance lock condition underpins the entire feedback process:

$$\pi e \varphi = \gamma^{-1} \zeta(3) \tag{2}$$

This equation, consistent with prior symbolic attractor lock models [2], marks the threshold at which Vesica phase compression becomes energetically stable. It anchors the formation of toroidal attractor shells and modulates local vacuum geometry.

At the reversal threshold R_c, the black hole horizon transitions from an absorptive surface to a resonant shell:

$$\Psi_H(t, r) = A_H e^{-\Gamma t} \cos(\omega_r r - \varphi_0) \tag{3}$$

Here, Γ is the decoherence decay constant, and $\omega_r \approx 0.244$ Hz, derived from recursive curvature harmonics.

4 Vacuum Feedback and Lambda Modulation

As field coherence builds, the surrounding vacuum responds dynamically. We model this with a local vacuum density function:

$$\rho_\Lambda(r, t) = \rho_0 + \Delta\rho \cos(\omega_r r - \theta t) \tag{4}$$

This behavior implies a breakdown of the scalar Λ paradigm. Instead, we define a dynamic recursive threshold:

$$\Lambda_{\text{VP}} = \int_0^{R_c} \lambda_n e^{-\alpha_n r} \, dr \tag{5}$$

If $\Lambda_{\text{VP}} > \Lambda_{\text{vac}}$, the event horizon enters a reversal regime, forming a Bragg-like mirror for dark energy flux.

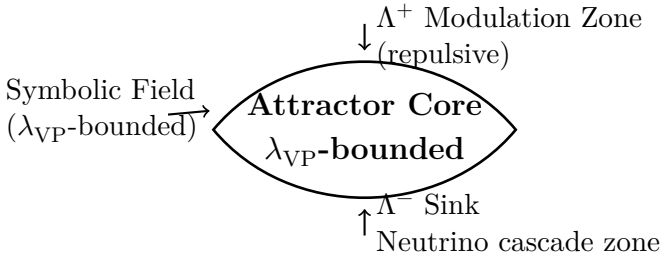


Figure 1: **Vesica Horizon Shell (Symbolic Lock)**

5 Recursive Cosmogenesis and Feedback Inflation

After horizon reversal, the attractor feeds entropy into internal inflation layers. Expansion follows:

$$R_n(t) = R_0 e^{\alpha_n t} \tag{6}$$

Recursive energy transfer between universes is described by:

$$\frac{dS_{\text{child}}}{dt} = -\frac{dS_{\text{parent}}}{dt} + \epsilon_f \tag{7}$$

This forms the feedback backbone for nested cosmogenesis and may account for spin biases observed on a galactic scale.

6 Conclusion

Dark energy is neither uniform nor passive. It is a resonance field encoded within the recursive attractor dynamics of the universe itself. Black holes reflect structured vacuum signals, and galactic-scale anomalies may stem from inherited symbolic curvature. The recursive Lambda modulation model offers a framework that unifies dark energy, quantum feedback, and Vesica-based field resonance.

References

[1] H. F. Halldórsson and ChatGPT. *Recursive Order Activation in Physical Systems*. July 2025.

[2] H. F. Halldórsson and ChatGPT. *Symbolic Attractors and Recursive Quantum Field Topologies*. 2025.

Appendix A: Mathematical Symbols and Physical Interpretation

$Q(t)$ **Quanta Field**
Non-local probabilistic field encoding standard particle behavior and conventional matter–energy dynamics.

$D(t)$ **Dark Matter Field**
Structured, phase-coherent mirror field orthogonal to $Q(t)$. Defined by a $\frac{\pi}{2\omega}$ phase shift:

$$D(t) = Q\Big(t + \tfrac{\pi}{2\omega}\Big). \tag{8}$$

φ **Golden Ratio** (≈ 1.618)
Used in Vesica field compression thresholds and recursive curvature harmonics.

$\pi e \varphi = \gamma^{-1}\zeta(3)$ **Resonance Lock Equation**
Symbolic attractor phase-lock threshold, encoding the onset of stable Vesica compression.

γ **Euler–Mascheroni Constant** (≈ 0.577)
Stabilizer coefficient in resonance-lock equations.

$\zeta(3)$ **Apéry's Constant** (≈ 1.202)
Governs recursive decay in symbolic energy modes.

ω_r **Resonant Horizon Frequency** ($\approx 0.244\,\mathrm{Hz}$)
Rotational frequency of recursive curvature modulation, tied to neutrino oscillations and teleportation pathways.

$\Psi_H(t, r)$ **Horizon Standing Wave Function**
Describes horizon reversal oscillation:

$$\Psi_H(t, r) = A_H\, e^{-\Gamma t}\, \cos(\omega_r\, r - \varphi_0). \tag{9}$$

Γ **Decoherence Decay Constant**
Exponential fading rate of recursive coherence at the horizon boundary.

$\rho_\Lambda(r, t)$ **Dynamic Vacuum Density Function**
Spatially modulated dark-energy field:

$$\rho_\Lambda(r, t) = \rho_0 \;+\; \Delta\rho\, \cos(\omega_r\, r - \theta\, t). \tag{10}$$

$\Lambda_{\mathbf{VP}}$ **Vesica Phase Lambda Threshold**
Dynamic field threshold for horizon reversal:

$$\Lambda_{\mathrm{VP}} = \int_0^{R_c} \lambda_n\, e^{-\alpha_n r}\, \mathrm{d}r. \tag{11}$$

λ_n **Recursive Field Harmonic**
The n-th decay mode of symbolic resonance.

α_n **Symbolic Field Decay Coefficient**
Spatial damping rate for harmonic n.

$R_n(t)$ **Recursive Expansion Function**
Internal inflation within a nested attractor:

$$R_n(t) = R_0\, e^{\alpha_n t}. \tag{12}$$

$\frac{\mathrm{d}S_{\text{child}}}{\mathrm{d}t} = -\frac{\mathrm{d}S_{\text{parent}}}{\mathrm{d}t} + \epsilon_f$ **Feedback Conservation Equation**
Entropy transfer between universes during recursive inflation; ϵ_f models leakage or Vesica boundary delay.

λ_{VP} **Vesica Phase Wavelength (Model-defined)**
Symbolic curvature wavelength, encoding field folding and glyph placement scale.

$\Delta r_n = \frac{2\pi}{n\,\omega_r}$ **Vesica Interference Node Spacing**
Resonance peaks in horizon structure; relevant for entanglement storage and curvature spike alignment.

Appendix B: Equation Index and Physical Purpose

(1) Dual-Field Phase Offset

$$D(t) = Q\left(t + \frac{\pi}{2\omega}\right)$$

Establishes the fundamental $\frac{\pi}{2}$ phase-shift between the quanta and dark matter fields. This orthogonality is the root of Vesica interference and recursive collapse geometry.

(2) Resonance Lock Condition

$$\pi e\varphi = \gamma^{-1}\zeta(3)$$

Symbolic attractor lock threshold — defines when Vesica compression becomes stable enough to initiate toroidal confinement and horizon structuring.

(3) Horizon Standing Wave Function

$$\Psi_H(t, r) = A_H e^{-\Gamma t}\cos(\omega_r r - \varphi_0)$$

Models the dynamic oscillation of a horizon transitioning from an absorptive boundary to a recursive resonator. Core to the reversal hypothesis.

(4) Vacuum Density Modulation Function

$$\rho_\Lambda(r, t) = \rho_0 + \Delta\rho\cos(\omega_r r - \theta t)$$

Describes how dark energy responds to recursive field oscillation. Replaces the static cosmological constant Λ with a dynamic feedback field.

(5) Lambda Crossover Threshold (Recursive)

$$\Lambda_{\text{VP}} = \int_0^{R_c} \lambda_n e^{-\alpha_n r}\, dr$$

Determines when recursive field energy exceeds vacuum stability — triggers horizon reversal and dark energy reflection behavior.

(6) Recursive Inflation Function

$$R_n(t) = R_0 e^{\alpha_n t}$$

Describes internal inflation driven by entropy-fed attractor growth after horizon reversal — each nested layer evolves from parent recursive curvature.

(7) Entropy Feedback Between Universes

$$\frac{dS_{\text{child}}}{dt} = -\frac{dS_{\text{parent}}}{dt} + \epsilon_f$$

Encodes the conservation and redistribution of entropy across recursive cosmogenesis. ϵ_f includes loss from boundary delay or Vesica asymmetry.

(Δ) Node Spacing from Vesica Interference

$$\Delta r_n = \frac{2\pi}{n\omega_r}$$

Defines radial spacing of constructive interference pockets along the horizon shell. These become sites for symbolic information storage, entanglement bubbles, or curvature anchoring.